

DEVELOPMENT OF A NEW 4-AMINO-1,8-NAPHTHALIMIDE DERIVATIVE AS A FLUORESCENT PROBE FOR MONITORING THE DIVALENT COPPER ION

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This article presents the design and synthesis of a new variant of the 4-amino-1,8-naphthalimide fluorescent probe for monitoring the divalent copper ion, Cu^{2+} . The probe BRST, functionalized with two propionic carboxylic acid groups in its receptor moiety, demonstrated high selectivity and sensitivity for the detection of Cu^{2+} in competitive media of various metal ions in DMSO/HEPES buffer solution (1:1; pH 7.4), with a fluorescence quenching of 75.41%. Besides its good linearity with Cu^{2+} , the probe yielded commendable values of the association constant and the detection limit, $5.54 \times 10^5 \text{ M}^{-1}$ and $4.33 \times 10^{-9} \text{ M}$, respectively. The quenching of the fluorescence intensity of the sensor in the presence of Cu^{2+} ions is attributable to the strong paramagnetic nature of the Cu^{2+} ion. The fluorescence “turn off” behavior of this simple and effective probe was successfully applied for the determination of Cu^{2+} levels in untreated water samples, with excellent recovery rates of 98–105.8%.

Keywords: 4-amino-1,8-naphthalimide, fluorescent probe, divalent copper ion, synthesis.

РАЗРАБОТКА НОВОГО ПРОИЗВОДНОГО 4-АМИНО-1,8-НАФТАЛИМИДА КАК ФЛЮОРЕСЦЕНТНОГО ЗОНДА ДЛЯ МОНИТОРИНГА ДВУХВАЛЕНТНОЙ МЕДИ

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Синтезировано новое производное 4-амино-1,8-нафталимида — флуоресцентный зонд для мониторинга ионов меди Cu^{2+} . Зонд BRST, функционализированный двумя пропионовыми карбоксильными кислотными группами в их рецепторной части, продемонстрировал высокую селективность и чувствительность к детектированию Cu^{2+} при наличии конкурентных сред (различных ионов металлов) в буферном растворе DMSO/HEPES (1:1, pH 7.4) с достигнутым тушением флуоресценции 75.41 %. Помимо линейности относительно концентрации ионов Cu^{2+} зонд обнаружил хорошие значения константы ассоциации и предела обнаружения: $5.54 \times 10^5 \text{ M}^{-1}$ и $4.33 \times 10^{-9} \text{ M}$. Тушение флуоресценции сенсора при определении Cu^{2+} связано с сильной парамагнитной природой данных ионов. Воспроизводимость результатов находится в пределах 98–105.8 %. Данный зонд может быть использован для обнаружения ионов Cu^{2+} в образцах воды.

Ключевые слова: 4-амино-1,8-нафталимид, флуоресцентный зонд, двухвалентный ион меди, синтез.

Introduction. The design of probes for metal ions is of significant interest in chemistry, biology, and environmental science [1–5]. Importantly, the fluorescence sensing of metal ions has emerged as an area of profound interest because it offers the relative advantages of high sensitivity and selectivity, simplicity of operation, low cost, and rapid analytical detection [6–9] over traditional techniques of detecting metal ions. Over a few decades, researchers have shifted focus to the monitoring of the divalent copper ion Cu^{2+} due to its environmental and biological roles which cannot be overestimated [10–15]. Cu^{2+} has a strong paramagnetic property (owing to its d^9 electronic configuration). It is known that this ion is a powerful fluorescence

emission quencher [16–20]. To date, considerable achievements have been realized in the design and synthesis of fluorescent probes for Cu^{2+} [21–30]. Meanwhile, there is ever increasing interest in the generation of more effective variants of Cu^{2+} -selective fluorescent probes for improved detection of Cu^{2+} in the competitive media of other environmentally and biologically important metal ions.

4-Amino-1,8-naphthalimide has been widely employed as the fluorophore moiety for the design of chemical probes. Researchers are motivated by its excellent photophysical fluorescent properties of high photostability and quantum yield and a strong absorption band in the visible region coupled with large Stokes shifts [31–34].

In this paper, we present the synthesis of a new variant of the 4-amino-1,8-naphthalimide probe as a fluorescent probe for the sensitive and selective monitoring of Cu^{2+} . The full characterization of the chemical structure and purity of the synthesized probe BRST appears in the experimental section of this paper. We have demonstrated that the probe BRST has outstanding sensing properties towards Cu^{2+} among the various competitive metal ions. The probe was eventually applied for the monitoring of Cu^{2+} in environmental water samples.

Experimental. Unless otherwise noted, our materials were obtained from commercial suppliers and used without further purification. Melting points were measured on a Melt-temp melting point electrothermal apparatus and remained uncorrected. The corrected excitation and fluorescence spectra were measured on a Perkin Elmer LS 55 fluorescence spectrophotometer. MS spectra were obtained on a microTOF mass spectrometer, while elemental analyses were determined using a Carlo Erba 1106 elemental analyzer. The IR spectra were recorded in the $4000\text{--}400\text{ cm}^{-1}$ region with a Nicolet FT-VERTEX 70 spectrometer using KBr pellets. Electronic spectra were taken on a Lab-Tech UV Bluestar spectrophotometer. ^1H NMR spectra were recorded on a Varian VR300-MHz spectrometer, and ^{13}C NMR spectra were obtained with a Mercury plus 400 MHz NMR spectrometer with TMS as the internal standard and $\text{DMSO-}d_6$ as the solvent. TLC was performed on silica gel, Fluka F60 254, 20×20 , 0.2 mm, using as the eluant the solvent systems ethyl acetate/dichloromethane (1:1). The synthetic route to chemosensor BRST is given in Fig 1.

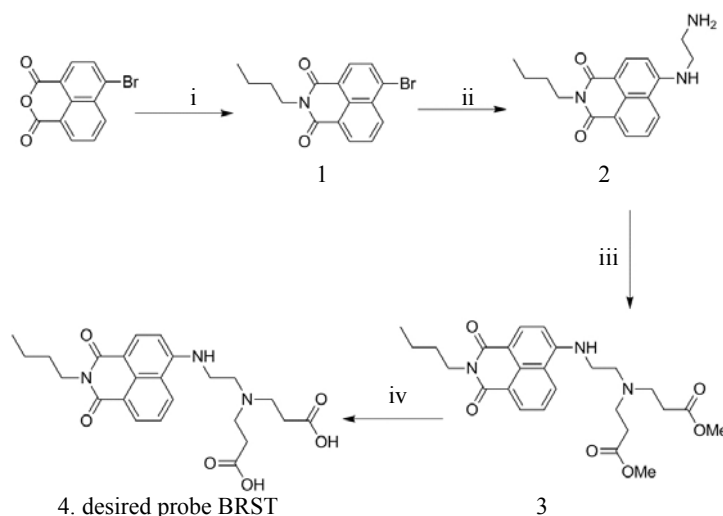


Fig. 1. Reagents and conditions for the synthesis of the BRST probe: i – $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$, EtOH; ii – $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$, iii – 2 molecules of methylacrylate ($\text{CH}_2\text{CHCOOCH}_3$), MeOH, iv – 3M KOH, MeOH, reflux 2–3 h, 75°C ; cation exchange chromatography (strong acid resin).

Procedures for spectroscopic detection of metal ions. Stock solutions of the metal ions (1 mM) were prepared from the metal salts NaNO_3 , KNO_3 , CaCl_2 , MgCl_2 , $\text{Al}(\text{NO}_3)_3$, $\text{Pb}(\text{NO}_3)_2$, $\text{Fe}(\text{NO}_3)_3$, $\text{Ni}(\text{NO}_3)_2$, $\text{Zn}(\text{NO}_3)_2$, $\text{Cu}(\text{NO}_3)_2$, $\text{Hg}(\text{NO}_3)_2$, AgNO_3 , $\text{Co}(\text{NO}_3)_2$, $\text{Cr}(\text{NO}_3)_3$, $\text{Mn}(\text{NO}_3)_2$, and $\text{Cd}(\text{NO}_3)_2$. Test solutions of the metal ions were prepared using double-distilled water by diluting the stock solutions to the desired molar concentration (0.1 mM). Similarly, the stock solution (1 mM) of the probe BRST was prepared in dimethylsulfoxide (DMSO) solvent, while the test solution of the probe was obtained by diluting the stock solution to the desired molar concentration (0.1 mM). The absorption and fluorescence spectra of the solution of the probe-metal ion were recorded using a mixture of 2 mL of HEPES (4-(2-hydroxyethyl)-1-piperazine ethanesulfonic acid) buffer solution, 20 μL of the sample solution of the probe, and 100 μL of each of the various

metal ions in a 1×1 cm quartz cuvette. The mixture was allowed to stand for 1 min for thorough mixing. The fluorescence measurements were carried out at the maximum excitation wavelength of 435 nm and the maximum emission wavelength of 750 nm, while the absorbance measurements were carried out in the wavelength range of 180–550 nm.

Synthesis of compound 1. A mixture of 4-bromo-1,8-naphthalic anhydride (15.00 g, 0.054 mol) and *n*-butylamine (7.90 g, 0.108 mol) in 300 ml of ethanol was heated under reflux with stirring for 3 h. The crude product that precipitated on cooling was filtered off, washed with ethanol, and dried to give 17.00 g (70%) of pure 4-bromo-*N*-*n*-butyl-1,8-naphthalimide (compound 1) as a light yellow solid. M.p. 107–109°C.

Synthesis of compound 2. A 500 mL round-bottom flask was charged with a solution of ethylenediamine (12.20 g, 201.67 mmol), and the reaction mixture was heated up to a temperature of 65°C. Thereafter, compound 1 (1.00 g, 3.01 mmol) was added to the portion above and the resulting mixture was heated to reflux for 3 h with stirring. After the reaction was complete, the reaction mixture was cooled to room temperature. Then 10 mL of distilled water was poured into the solution, which resulted in the formation of a yellow precipitate. The precipitate formed was isolated by filtration and recrystallized from toluene to afford yellow-orange crystals of 4-(2-aminoethyl)amino-*N*-*n*-butyl-1,8-naphthalimide (compound 2) in 72.40 % yield. M.p. 132–133°C.

Synthesis of compound 3. To a solution of compound 2 (2.00 g, 6.40 mmol) in 80 mL of methanol was added slowly a solution of methylacrylate (5.50 g, 64 mmol) in 50 ml of methanol via a syringe for a period of 30 min. The reaction mixture was stirred for 12 days at room temperature, and the excess of methylacrylate was removed by suction filtration, whereupon the ester-functionalized derivative 3 was obtained as a yellow-brown solid in 98.21% yield and then dried in vacuum.

Synthesis of compound 4 (the desired probe BRST). The desired compound BRST was obtained by passing compound 3 through a strong cation-exchange chromatographic resin. The solution containing the desired compound that ran down from the column was heated off indirectly on a water condenser to afford the desired probe BRST in 60.87% yield. The crude product was finally recrystallized in water to give the pure product in 77.04% yield. M.p: 217–219°C. MS (ES⁺) m/z = 456.2067 (M + H)⁺. FT-IR ($\nu_{\max}$, KBr, cm⁻¹): 759 (ν_{Ar}); 3381 (ν_{NH}); 2603 and 2828 (ν_N); 2871 and 2960 (ν_{CH}); 1684 ($\nu^{as}N-C=O$); 1646 ($\nu^sN-C=O$); 1717 ($\nu_{C=O}$ of COOH); 3663 and 3669 (ν_{OH} of COOH); 1550 and 1588 ($\nu_{C=C}$, aromatic); 1300 and 1362 (ν_{C-O} of COOH). UV-vis. (in DMSO, nm): 260, 444. ¹H NMR (300 MHz, DMSO-*d*₆): δ = 8.57 (d, 1H, J = 7.3 Hz, naphthalimide H-5); 8.45 (d, 1H, J = 8.6 Hz, naphthalimide H-2); 8.38 (dd, 1H, J = 8.3 Hz, J = 1.0 Hz, naphthalimide H-7); 7.60 (dd, 1H, J = 8.3 Hz, J = 7.3 Hz, naphthalimide H-6); 6.67 (d, 1H, J = 8.6 Hz, naphthalimide H-3); 6.28 (m, 1H, NH); 4.16 (t, 2H, J = 7.6 Hz, (CO)₂NCH₂); 11.34 (s, 2H, 2×OH); 3.44 (m, 2H, ArNHCH₂); 2.83 (m, 6H, 3×NCH₂); 2.34 (t, 4H, J = 6.4 Hz, 2×CH₂COOH); 1.57 (m, 2H, CH₂CH₂CH₂CH₃); 1.32 (m, 2H, CH₂CH₂CH₂CH₃); 0.96 (t, 3H, J = 7.2 Hz, CH₂CH₂CH₂CH₃). ¹³C NMR (400 MHz, DMSO-*d*₆): 178.83, 168.79, 167.96, 155.42, 139.19, 139.19, 135.63, 134.37, 133.29, 132.75–131.72, 129.38, 126.92, 125.17, 112.99, 108.94, 56.16, 54.05, 45.83, 45.23, 45.02, 44.81, 44.60, 44.40, 44.19, 44.05, 37.11, 34.95, 25.00, 18.88.

Results and discussion. The route to the synthesis of the probe BRST is explicitly illustrated in Fig. 1. 4-Bromo-1,8-naphthalic anhydride was utilized as the starting material. Intermediate compounds 1–3 were prepared following the procedures reported by Bojinov et al. [3]. The full characterization of the probe was done by Fourier transform infrared red (FT-IR), ultraviolet-visible (UV-vis), proton nuclear magnetic resonance (¹H NMR) and carbon-13 nuclear magnetic resonance (¹³C NMR) spectroscopic methods, and MALDI-HRMS spectrometry¹. The probe obtained displayed outstanding solubility in dimethylformamide (DMF) and dimethylsulfoxide (DMSO), good solubility in ethanol, methanol, and tetrahydrofuran, partial solubility in water, acetone, and acetonitrile; and total insolubility in dichloromethane and trichloromethane.

Photophysical properties. This investigation assesses the photophysical characteristics (absorption maxima, λ_A ; fluorescence maxima, λ_F ; Stokes shift, $\nu_A - \nu_F$; and quantum fluorescence yield, Φ_F) of the probe BRST in five solvents of varying polarities. As depicted in Table 1, it was found that increasing the solvent polarity led to an increase in the bathochromic shift of the longest wavelength band. DMSO, the solvent having the highest polarity, gave the highest values of λ_A and λ_F , while tetrahydrofuran, the solvent with the lowest polarity, yielded the lowest values of λ_A and λ_F . Next obtained were the values of the Stoke's shift, which is a photophysical parameter of fluorescent compounds that signifies the differences in the properties

¹ ¹H and ¹³C NMR spectrum of BRST and HRMS spectrum of BRST and [Cu(NO₃)₂ + BRST] is available upon request from authors.

and structure of the fluorophores between the ground state S_0 and the first excited state S_1 and is calculated using the equation [35]

$$\nu_A - \nu_F = (1/\lambda_A - 1/\lambda_F) \times 10^7 \text{ cm}^{-1}. \quad (1)$$

The calculated Stoke's shift values were obtained within the range 3368.1–4152.9 cm^{-1} . More so, the values of the molar extinction coefficient are in the range 39030–700700 $\text{L} \times \text{mol}^{-1} \times \text{cm}^{-1}$, which is in accordance with those of similar 1,8-naphthalimide derivatives [36–38]. The ability of molecules to emit the absorbed light is characterized quantitatively by the fluorescence quantum yield (Φ_F), which was determined using the equation [36], with *N*-butyl-4-*n*-butylaminonaphthalimide ($\Phi_F = 0.81$ in ethanol) as the standard:

$$\Phi_F = \Phi_{\text{ref}}(S_{\text{sml}}/S_{\text{ref}})(A_{\text{ref}}/A_{\text{sml}})(n_{\text{sml}}/n_{\text{ref}})^2, \quad (2)$$

where Φ_F and Φ_{ref} are the emission quantum yield of the sample and standard, A_{ref} and A_{sml} are the absorbance of the standard and sample at their respective excited wavelength, S_{ref} and S_{sml} denote the integrated emission band areas of the standard and sample, respectively, and n_{ref} and n_{sml} represent the solvent refractive index of the standard and sample, respectively. Noteworthy is the decrease in the value of the quantum yield with increasing the solvent polarity, with DMSO and DMF having the lowest values and THF having the highest. This could be attributed to the PET quenching phenomenon usually observed in nonpolar solvents, which in turn leads to the regeneration of the fluorescence emission [39].

TABLE 1. Summary of the Photophysical Properties of the Proposed BRST Probe in Solvents of Varying Polarities

Solvent	λ_F , nm	λ_A , nm	ϵ_{max} , $10^4 \text{ L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$	$\nu_A - \nu_F$, cm^{-1}	Φ_F
DMSO	528	444	3.90	3368.1	0.18
DMF	521	440	70.07	3533.4	0.02
THF	504	428	4.51	3562.5	0.96
EtOH	521	436	4.59	3705.0	0.26
MeOH	524	429	3.93	4152.9	0.40

pH sensing properties. Generally, in proximity to fluorescent probes, protons influence the sensing abilities and the quality of the monitoring of an analyte [40]. pH titration experiments of the probe BRST were conducted with solutions of different pH values (0.54–13.58) adjusted with 0.1 M HCl and/or 0.1 M NaOH. As shown in Fig. 2a, upon acidification, only minor changes (<3.51% variation) in the fluorescence intensity of the probe occurred from the starting pH of 0.54 to 8.62. In other words, the determination of Cu^{2+} with the probe BRST was not significantly affected by changes in the pH in this region. This validates that the designed probe could facilitate the monitoring of Cu^{2+} in a wide pH span and behave as a functional probe in acidic and medium pH value regions. Based on the experimental results as justified in Fig. 2b, pH 7.4 was selected as the optimum pH in which the probe works best.

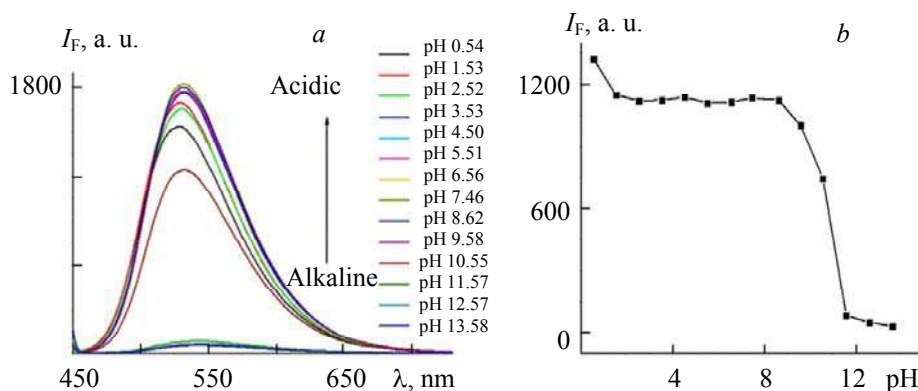


Fig. 2. (a) pH effect on the fluorescence spectra of the BRST probe (0.1 mM) in DMSO solution. The pH was modulated by adding 0.1 M HCl and/or 0.1 M NaOH in double-distilled water (b). The changes at 529 nm as a function of pH.

Fluorescence titration experiments. The fluorescence titration reaction curve of the changes in the fluorescence spectra of the probe towards Cu^{2+} was investigated. Figure 3a shows the maximal fluorescence band of the BRST probe, which is centered at 532 nm. As anticipated, the fluorescence intensity of the probe was quenched steadily in response to the gradual increase in the concentrations of the added Cu^{2+} in the DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) with observed bathochromism of about 3 nm. This observation could be explained on the basis of two reasons: the complexation of Cu^{2+} that reduces the electron-releasing ability of the nitrogen atom in the receptor moiety, thereby inhibiting the photoinduced electron transfer (PET) process, and the paramagnetic property of Cu^{2+} . As Fig. 3b clearly reveals, the concentration-dependent fluorescence quenching of the probe decreased linearly as a function of increasing Cu^{2+} , even up to when 50 μM of Cu^{2+} solution was added.

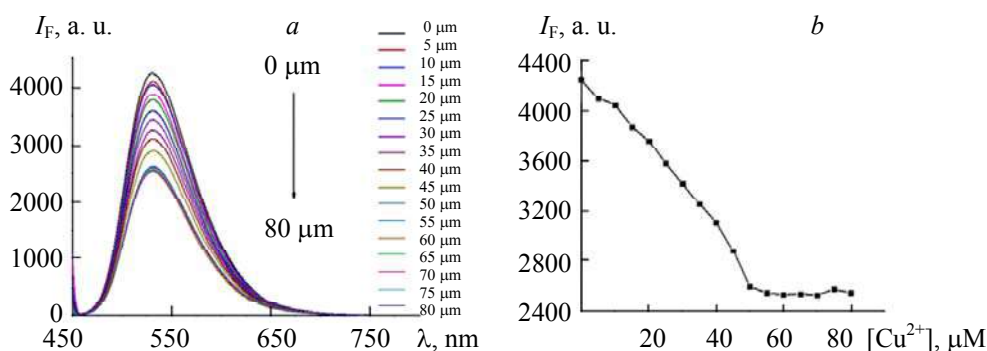


Fig. 3. (a) Changes in the fluorescent spectra of the BRST probe (0.1 mM) in DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) upon the addition of different concentrations of Cu^{2+} (0.04 mM). (b) The changes in the fluorescence intensity at 529 nm as a function of Cu^{2+} concentrations ($\lambda_{\text{exc}} = 435 \text{ nm}$).

Identification property and selectivity study of the probe BRST. The BRST probe displays excellent solubility in DMSO, so its spectra were obtained in the DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4). The response of the probe to various metal ions was investigated under excitation at 435 nm. When no metal ion was present, the probe exhibited a high fluorescence peak (a sharp green fluorescence) at 532 nm. The introduction of the metal ions Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Al^{3+} , Pb^{2+} , Fe^{3+} , Ni^{2+} , Zn^{2+} , Hg^{2+} , Ag^+ , Co^{2+} , Cr^{2+} , Mn^{2+} , Cd^{2+} , and Cu^{2+} into the 0.1 mM DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) of the probe led to the fluorescence of the probe being quenched by up to 75.41% (which represents a 4.07-fold decrease in the fluorescence intensity) only by Cu^{2+} (Fig. 4). This is ascribable to the powerful paramagnetic nature of Cu^{2+} [16–20]. In contrast, the other metal ions investigated under the same conditions had negligible effects on the fluorescence intensity of the probe. Thus, the proposed probe seems useful for the selective monitoring of Cu^{2+} even when in coexistence with other metal ions.

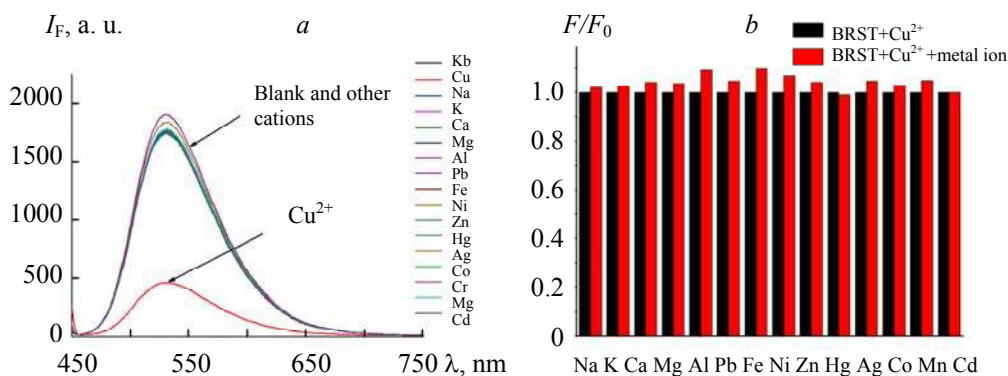


Fig. 4. (a) Fluorescence spectra of the BRST probe (0.1 mM) in DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) in the presence of different amounts of competing metal ions (0.1 mM). (b) Fluorescence response of the BRST probe (0.1 mM) to Cu^{2+} (0.1 mM) or to a mixture of each of the specified metal ions (0.1 mM) with Cu^{2+} (0.1 mM) in DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4).

Calculation of association constant and detection limit. The association constant and detection limit of the BRST probe were determined based on the 1:1 stoichiometry. First, the association constant (K_a) was estimated using the Benesi-Hildebrand equation [41]:

$$1/(F - F_0) = 1/K_a(F_{\min} - F_0)[\text{Cu}^{2+}]^n + 1/(F_{\min} - F_0), \quad (3)$$

where F is the fluorescence intensity at 529 nm at any given concentration of Cu^{2+} , F_0 is the fluorescence intensity at 529 nm in the absence of Cu^{2+} , $F_{\min}$ is the minimal fluorescence intensity at 529 nm in the presence of Cu^{2+} in the solution, and n is the stoichiometric mole ratio, which is 1 in this case. The association constant was evaluated graphically by plotting $1/(F - F_0)$ against $1/[\text{Cu}^{2+}]$. Data were linearly fitted according to the above equation, and the K_a value was obtained from the slope of the line. The value of the association constant was calculated to be $5.54 \times 10^5 \text{ M}^{-1}$, which is large enough and indicates a strong binding interaction between the BRST probe and Cu^{2+} .

The detection limit of BRST towards Cu^{2+} was evaluated based on the equation $\text{DL} = 3\sigma/k_{\text{SV}}$, where σ is the standard deviation of the blank solution and k_{SV} is the slope from the calibration plot. The value of k_{SV} is obtained from the Stern–Volmer plot of the ratio of the fluorescence intensities versus the concentration of Cu^{2+} based on the equation [42]:

$$F_0/F = 1 + K_{\text{SV}}[Q], \quad (4)$$

where F_0 is the initial fluorescence intensity, F is the fluorescence intensity measured in the presence of the analyte, $[Q]$ is the concentration of the quencher, which is $[\text{Cu}^{2+}]$ in this case, and K_{SV} is the Stern–Volmer constant. The plot of (F_0/F) against $[\text{Cu}^{2+}]$ gave a straight line, with the value of the slope equal to K_{SV} . The obtained value of the detection limit was $4.33 \times 10^{-9} \text{ M}$, a much lower value than the WHO and U.S. EPA regulated limits of 35 and 20 μM . The low value of the detection limit shows the capability of the probe to monitor minute concentrations of Cu^{2+} , whether in biological or environmental systems.

The Job method of continuous variation was used to obtain the coordination stoichiometry of the complex formed [43], $[\text{Cu}(\text{BRST})]$, between the BRST probe and Cu^{2+} . The maxima at the mole fraction 0.5 of the Job plot (Fig. S1) clearly reveals that Cu^{2+} forms a 1:1 complex with the BRST probe. The association constant (Fig. S2) and detection limit of the BRST probe (Fig. S3) were determined based on the 1:1 stoichiometry to be $5.54 \times 10^5 \text{ M}^{-1}$ and $4.33 \times 10^{-9} \text{ M}$.

Analytical application of probe BRST. The proposed BRST probe was used to determine the amounts of Cu^{2+} in real water samples obtained from the Yellow River (Lanzhou city) and tap water from our school, Lanzhou Jiaotong University. The pH of these water samples were suitably adjusted by the DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) and analyzed with the BRST probe. Table 2 shows the results of the analysis. The high values of recoveries of spiked Cu^{2+} obtained (98–105.8%) indicate the potential of the synthesized probe to track Cu^{2+} in real water samples.

TABLE 2. Results of the Water Sample Analysis for Divalent Copper Ion Determination Using the Proposed BRST Probe

Real water samples	Cu^{2+} added, μM	Cu^{2+} recovery $\pm$ mean SD, μM	Recovery, %
Yellow River	2	2.02 $\pm$ 0.13	101
	3	3.14 $\pm$ 0.20	104.7
	4	3.98 $\pm$ 0.16	99.5
Tap	2	1.96 $\pm$ 0.03	98
	3	3.07 $\pm$ 0.25	102.3
	4	4.23 $\pm$ 0.32	105.8

Conclusion. A new 4-amino-1,8-naphthalimide derivative was designed and synthesized as a fluorescent probe for monitoring the divalent copper ion Cu^{2+} . The results illustrate that the BRST probe displayed an unalloyed quenching property towards Cu^{2+} in the DMSO/HEPES buffer solution (v/v, 1:1; pH 7.4) in the presence of other competitive metal ions such as Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Al^{3+} , Pb^{2+} , Fe^{3+} , Ni^{2+} , Zn^{2+} , Hg^+ , Ag^+ , Co^{2+} , Cr^{2+} , Mn^{2+} , and Cd^{2+} . Moreover, the detection limit of the probe for the monitoring of Cu^{2+} reached the level of $4.33 \times 10^{-9} \text{ M}$, making the probe highly utilizable for the quantitative analysis of Cu^{2+} in both environmental and biological systems.

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Supplementary

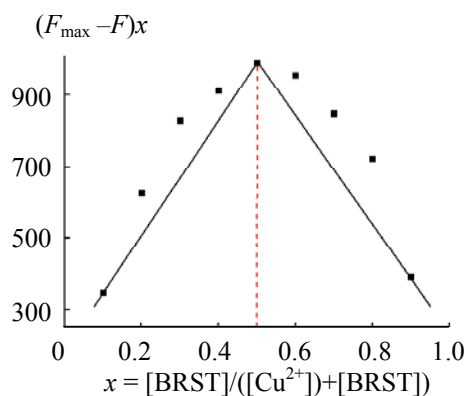


Fig. S1. Job plot for the complex $[\text{Cu}(\text{BRST})]$ (the total concentration of BRST and Cu^{2+} was kept at 100 μM).

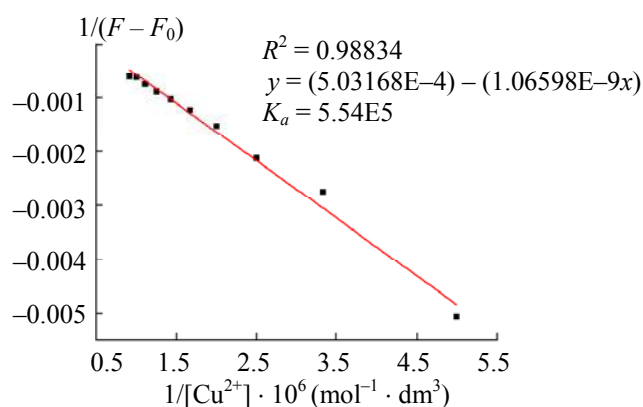


Fig. S2. Benesi-Hildebrand plot for the determination of the association constant of complex $[\text{Cu}(\text{BRST})]$, assuming 1:1 stoichiometry for the association between BRST and Cu^{2+} .

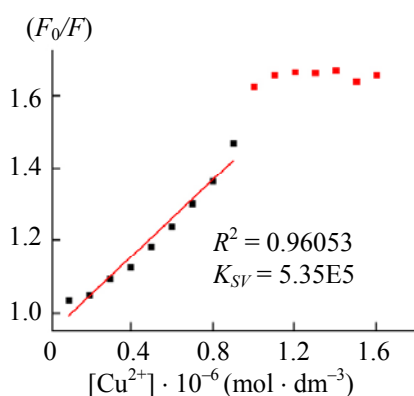


Fig. S3. Stern-Volmer plot of the ratio of fluorescence intensities F_0/F (where F_0 is the fluorescence intensity of the blank, equal to 4245) of the BRST probe against increasing concentration of Cu^{2+} ($[\text{Cu}^{2+}] = 5\text{--}80 \mu\text{M}$).

REFERENCES

1. N. Kaur, S. Kumar, *Tetrahedron*, **67**, 9233–9264 (2011).
2. A. Caballero, R. Martínez, V. Lloveras, I. Ratera, J. Vidal-Gancedo, K. Wurst, A. Tárraga, P. Molina, J. Veciana, *J. Am. Chem. Soc.*, **127**, 15666–15667 (2005).

3. V. B. Bojinov, N. I. Georgiev, P. S. Nikolov, *J. Photochem. Photobiol. A*, **193**, 129–138 (2008).
4. W. K. Dong, S. F. Akogun, Y. Zhang, Y. X. Sun, X. Y. Dong, *Sensor. Actuat. B: Chem.*, **238**, 723–734 (2017).
5. J. H. Hu, Y. Sun, J. Qi, Q. Li, T. B. Wei, *Spectrochim. Acta A*, **175**, 125–133 (2017).
6. F. Wang, Y. L. Xu, S. O. Aderinto, H. P. Peng, H. Zhang, H. L. Wu, *J. Photochem. Photobiol. A*, **332**, 273–282 (2017).
7. W. K. Dong, X. L. Li, L. Wang, Y. Zhang, Y. J. Ding, *Sensor. Actuat. B: Chem.*, **229**, 370–378 (2016).
8. Y. L. Xu, S. S. Mao, H. P. Peng, F. Wang, H. Zhang, S. O. Aderinto, H. L. Wu, *J. Lumin.*, **192**, 56–63 (2017).
9. S. Thavornpradit, J. Sirirak, N. Wanichacheva, *J. Photochem. Photobiol. A*, **330**, 55–63 (2016).
10. C. Barranguet, F. P. van den Ende, M. Rutgers, A. M. Breure, M. Greijdanus, J. J. Sinke, W. Admiraal, *Environ. Toxicol. Chem.*, **22**, 1340–1349 (2003).
11. J. Liu, Y. Lu, *J. Am. Chem. Soc.*, **129**, 9838–9839 (2007).
12. Y. Sun, J. H. Hu, J. Qi, J. B. Li, *Spectrochim. Acta A*, **167**, 101–105 (2016).
13. R. Kramer, *Angew. Chem. Int. Edit.*, **37**, 772–773 (1998).
14. G. L. Millhauser, *Acc. Chem. Res.*, **37**, 79–85 (2004).
15. E. Gaggelli, H. Kozłowski, D. Valensin, G. Valensin, *Chem. Rev.*, **106**, 1995–2044 (2006).
16. W. Y. Lin, L. Yuan, W. Tan, J. B. Feng, L. L. Long, *Chem. Eur. J.*, **15**, 1030–1035 (2009).
17. R. Bergonzi, L. Fabbrizzi, M. Licchelli, C. Mangano, *Coord. Chem. Rev.*, **170**, 31–46 (1998).
18. Q. T. Meng, Y. Shi, C. P. Wang, H. M. Jia, X. Gao, R. Zhang, Y. F. Wang, Z. Q. Zhang, *Org. Biomol. Chem.*, **13**, 2918–2926 (2015).
19. R. Zhang, X. J. Yu, Y. J. Yin, Z. Q. Ye, G. L. Wang, J. L. Yuan, *Anal. Chim. Acta*, **691**, 83–88 (2011).
20. M. H. Lim, B. A. Wong, W. H. Pitcock, Jr., D. Mokshagundam, M. H. Baik, S. J. Lippard, *J. Am. Chem. Soc.*, **128**, 14364–14373 (2006).
21. E. B. Veale, J. A. Kitchen, T. Gunnlaugsson, *Supramol. Chem.*, **25**, 101–108 (2013).
22. Z. Xu, J. Yoon, D. R. Spring, *Chem. Commun.*, **46**, 2563–2565 (2010).
23. N. Singh, N. Kaur, B. McCaughan, J. F. Callan, *Tetrahedron Lett.*, **51**, 3385–3387 (2010).
24. J. Huang, Y. Xu, X. Qian, *Dalton Trans.*, 1761–1766 (2009).
25. H. L. Wu, S. O. Aderinto, Y. L. Xu, H. Zhang, X. Y. Fan, *J. Appl. Spectrosc.*, **84**, 25–30 (2017).
26. E. L. Que, D. W. Domaille, C. J. Chang, *Chem. Rev.*, **108**, 1517–1549 (2008).
27. Y. L. Xu, S. O. Aderinto, H. L. Wu, H. P. Peng, H. Zhang, J. W. Zhang, X. Y. Fan, *Z. Naturforsch. B*, **72**, 35–41 (2017).
28. S. O. Aderinto, Y. L. Xu, H. P. Peng, F. Wang, H. L. Wu, X. Y. Fan, *J. Fluoresc.*, **27**, 79–87 (2017).
29. Y. Gao, Y. Li, X. Yang, F. He, J. Huang, M. Jiang, Z. Zhou, H. Chen, *RSC Adv.*, **5**, 80110–80117 (2015).
30. Y. Zhang, X. Guo, X. Tian, A. Liu, L. Jia, *Sensor. Actuat. B: Chem.*, **218**, 37–41 (2015).
31. H. L. Wu, H. P. Peng, F. Wang, H. Zhang, C. G. Chen, J. W. Zhang, Z. H. Yang, *J. Appl. Spectrosc.*, **83**, 931–937 (2017).
32. J. Wang, Y. Xiao, Z. Zhang, X. Qian, Y. Yang, Q. Xu, *J. Mater. Chem.*, **15**, 2836–2839 (2005).
33. S. O. Aderinto, H. Zhang, H. L. Wu, C. Y. Chen, J. W. Zhang, H. P. Peng, Z. H. Yang, F. Wang, *Color. Technol.*, **133**, 40–49 (2017).
34. H. L. Wu, C. Y. Chen, H. Zhang, H. P. Peng, F. Wang, Z. H. Yang, J. W. Zhang, *Chem. Pap.*, **70**, 685–694 (2016).
35. N. I. Georgiev, V. B. Bojinov, N. Marinova, *Sensor. Actuat. B: Chem.*, **150**, 655–666 (2010).
36. D. Staneva, I. Grabchev, J.-P. Soumillion, V. Bojinov, *J. Photochem. Photobiol. A*, **189**, 192–197 (2007).
37. H. P. Peng, K. S. Shen, S. S. Mao, X. K. Shi, Y. L. Xu, S. O. Aderinto, H. L. Wu, *J. Fluoresc.*, **27**, 1191–1200 (2017).
38. H. L. Wu, C. P. Wang, J. W. Zhang, Y. H. Zhang, C. Y. Chen, Z. H. Yang, X. Y. Fan, *Z. Naturforsch. B*, **70**, 863–869 (2015).
39. X. Poteau, A. I. Brown, R. G. Brown, C. Holmes, D. Matthew, *Dyes Pigments*, **47**, 91–105 (2000).
40. X. Sun, G. Kim, Y. Xu, J. Yoon, T. D. James, *Chem. Plus Chem.*, **81**, 30–34 (2016).
41. J. R. Lakowicz, *Principles of Fluorescence Spectroscopy*, Plenum Press, New York (1999).
42. O. Stern, M. Volmer, *Z. Phys.*, **20**, 183 (1919).
43. P. Job, *Ann. Chim. Appl.*, **9**, 113–203 (1928).